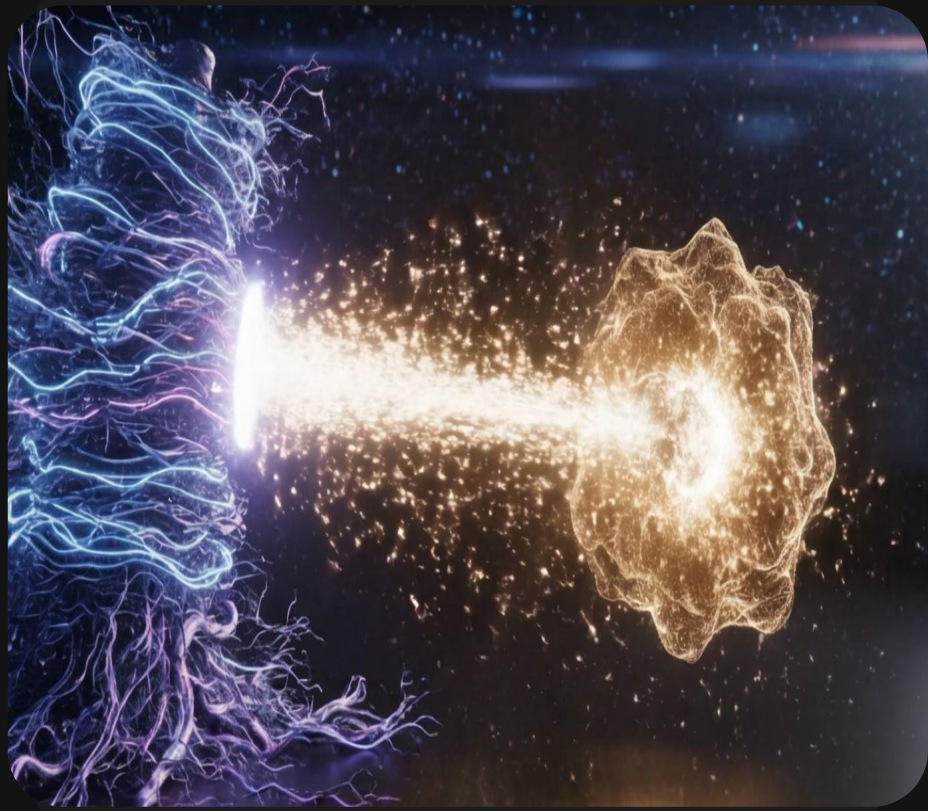
GOOGLE CLOUD - 2026

Unlocking TPU Performance Through Model Distillation

Join us in San Francisco to discover how Model Distillation compresses massive AI models into high-throughput 'students' custom-engineered to fully saturate the Matrix Multiply Unit (MXU) of Google TPU. By aligning your model architecture perfectly with TPU hardware, you will maximize processing utilization and unlock elite, cost-efficient serving performance at scale.

November 19, 2026, 3:00 – 5:00 PM PST

Google Office, 345 SPEAR St, San francisco, 94105, US



Agenda

Proprietary + Confidential

Agenda

01 Introduction

02 TPU Architecture & Model Architecture

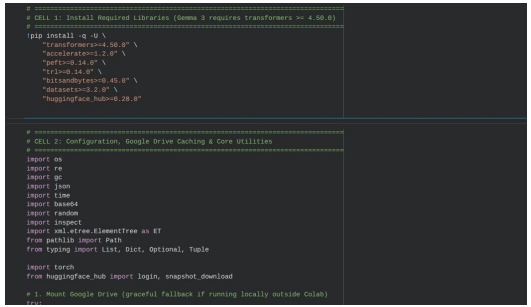
03 Distillation via 2 exemples

04 Next Steps

05 Q&A

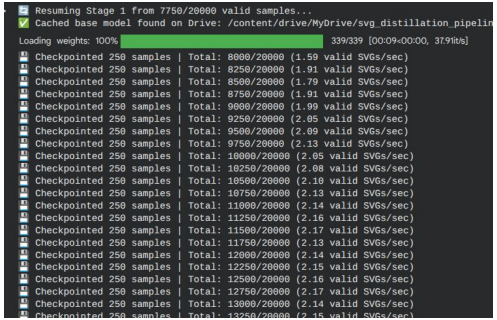

Qwen Coder 7B

colab

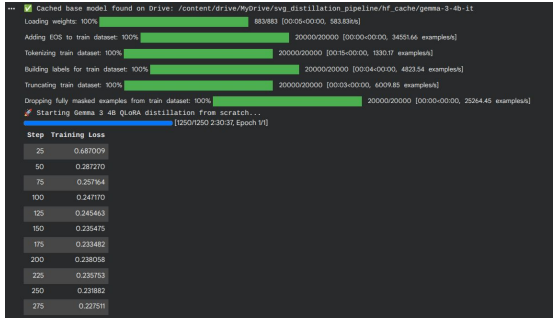


Setup & imports
Mount Drive

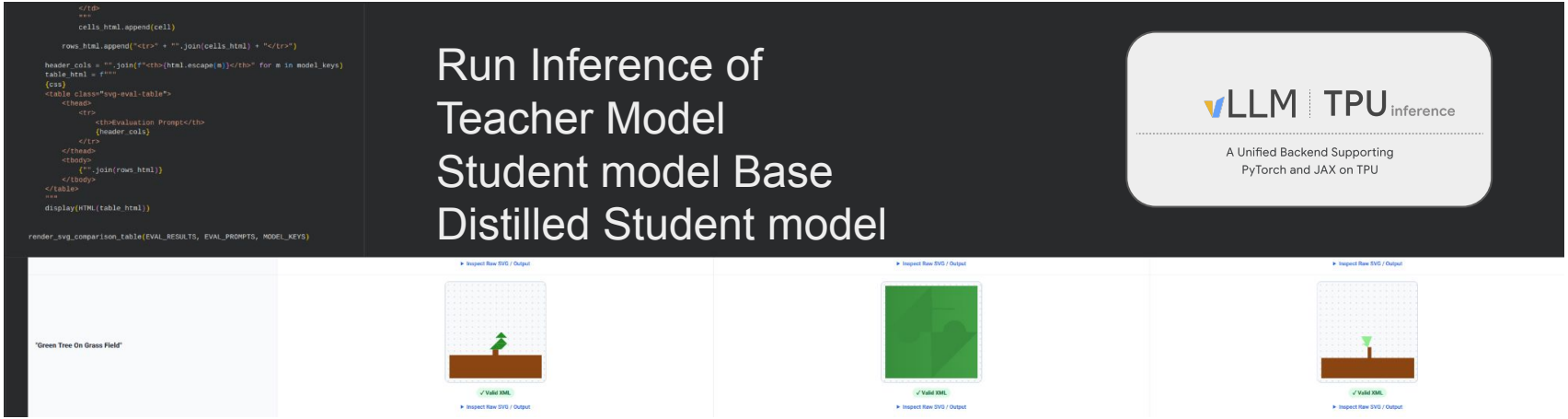
Google Drive



Generate Dataset
Qwen Coder 7B



Distill Gemma 3 4B



Run Inference of
Teacher Model
Student model Base
Distilled Student model

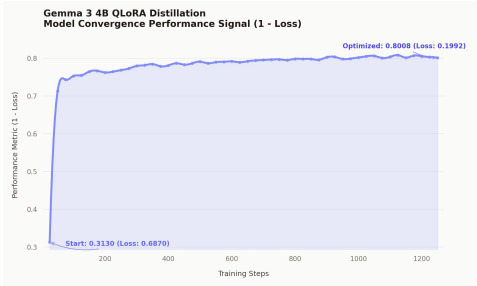
vLLM TPU inference

A Unified Backend Supporting
PyTorch and JAX on TPU

Qwen Qwen2.5-Coder

	Qwen2.5 Coder 32B Base	Qwen2.5 Coder 14B Base	Qwen2.5 Coder 7B Base	Qwen2.5 Coder 3B Base	Qwen2.5 Coder 1.5B Base	Qwen2.5 Coder 0.5B Base
HumanEval	**65.9**	64.0	61.6	52.4	43.9	28.0
MBPP-3shot	**76.4**	71.4	68.8	65.2	59.2	40.4
MultiPLE average	**63.9**	59.9	57.5	48.0	41.1	24.7
CRUXEval-O Cot	**69.4**	66.4	56.0	43.8	34.6	23.0
BigCodeBench Original Average	**40.0**	37.0	31.0	26.3	22.1	10.4
Fill-in-the-Middle average (Python, Java, JS)	**88.3**	87.7	86.2	85.7	83.5	77.7
GSM8K 8-shot	**91.1**	88.7	83.9	73.7	65.8	34.5
MATH 4-shot	**57.2**	52.8	46.6	40.0	30.9	15.4
MMLU	**79.1**	75.2	68.0	61.2	53.6	42.0
ARC-Challenge	**70.5**	66.0	60.9	52.9	45.2	34.4

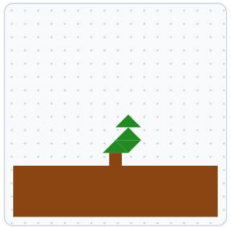
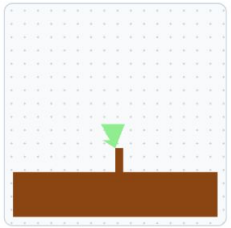
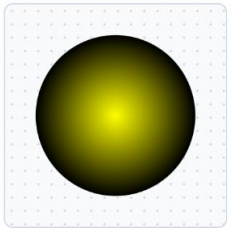
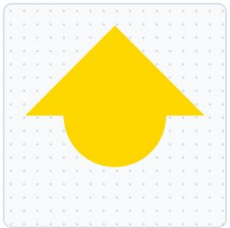
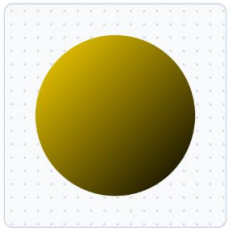
Distill

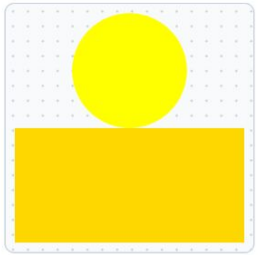
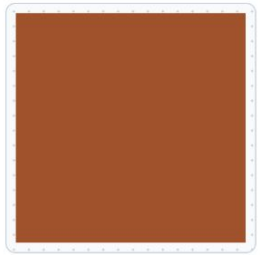
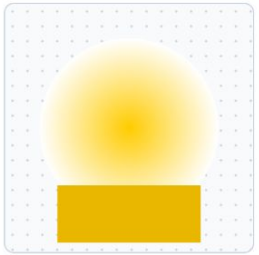
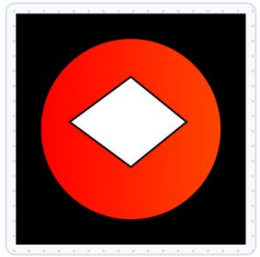
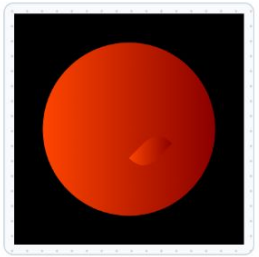


Distill

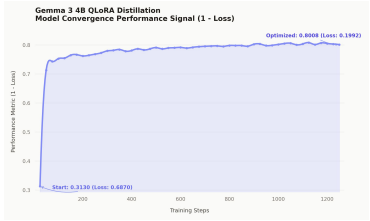
Gemma 3

	Gemma 3 larger	Gemma 3 larger	Gemma 3 large	Gemma 3 4 B	Gemma 3 smaller	Gemma 3 smallest
HumanEval	**65.9**	64.0	61.6	52.4	43.9	28.0
MBPP-3shot	**76.4**	71.4	68.8	65.2	59.2	40.4
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	Qwen2.5-Coder7B	Gemma 3 4B Base	Gemma 3 4B Distill
"Green Tree On Grass Field"	 ✓ Valid XML	 ✓ Valid XML	 ✓ Valid XML
"Yellow Star In Dark Night"	 ✓ Valid XML	 ✓ Valid XML	 ✓ Valid XML

	Qwen2.5-Coder7B	Gemma 3 4B Base	Gemma 3 4B Distill
"Yellow Sun Over Golden Beach"	 ✓ Valid XML	 ✓ Valid XML	 ✓ Valid XML
"Red Rocket In Deep Space"	 ✓ Valid XML ▶ Inspect Raw SVG / Output	⚠ Malformed or Missing <svg> XML ✗ Invalid XML ▶ Inspect Raw SVG / Output	 ✓ Valid XML ▶ Inspect Raw SVG / Output

deepseek



RL
GPRO

Gemma

Trillium TPU

15.8 x Cheaper per correct answer
than the teacher model

Google Antigravity

TPU-Distil: strong teacher, small TPU student

Distil DeepSeek-V4.1-Flash into Qwen3-1.7B, reinforce it with GRPO, and serve it on one Cloud TPU v6e chip

Text-to-SQL (BIRD dev, 1,534 questions) · financial QA (FinQA test, 1,147 questions) · same-run paired evaluation

+18 / +23	+4.0 / +5.8	16–17×	$136 / $79
points from distillation	more points from GRPO	cheaper per correct answer	to produce the student
BIRD / FinQA, final runs	BIRD / FinQA, 95% CIs above 0	1 v6e chip vs 8× H200	BIRD / FinQA, list prices

Google Cloud

±2	0	95%	2	1:1
pt serving noise floor	eval leaks after audit	CIs on every comparison	task families, one recipe	bit-identical retrain

The recipe, end to end

1 Teacher answers
33,728
DeepSeek-V4.1-Flash on 8× H200
4 samples per BIRD question
non-thinking mode · 38 min

check →

2 Verified data
10,112
only answers the task verifies
≤ 4 distinct per question
0 evaluation leaks (audited)

SFT →

3 Distil (KD)
27 min
Qwen3-1.7B on TPU v6e-8
117 steps · batch 256 · LoRA r256
Tunix · qwix · Optax · JAX

reward →

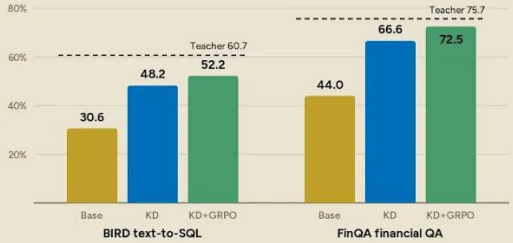
4 Reinforce (GRPO)
6 rounds
8 samples per question at T = 1
PPO clip 0.2 · KL to KD, beta 0.02
~45 min per round on v6e-8

merge →

5 Serve
15.8 req/s
vLLM-TPU on one v6e chip
$0.091 per 1,000 correct answers
15.8× cheaper than the teacher

Accuracy ladder

Final same-run evaluation per task · dashed line = teacher



Better and cheaper

BIRD serving cost per 1,000 correct answers · accuracy under each name



15.8× cheaper per correct answer than the teacher, and accuracy rises at every training stage

Where the gain comes from

KD gain over base, split by the self-distillation control

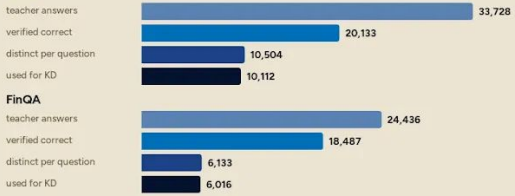


Paired differences, 95% CI (KD rows: control runs)

BIRD	distilled vs base	+18.1	[+15.6, +20.6]
BIRD	distilled vs self-distilled	+13.1	[+10.7, +15.5]
BIRD	KD + GRPO vs distilled	+4.0	[+2.2, +5.9]
FinQA	distilled vs base	+24.1	[+21.0, +27.2]
FinQA	distilled vs self-distilled	+9.6	[+6.9, +12.4]
FinQA	KD + GRPO vs distilled	+5.8	[+3.9, +7.8]

Teacher data funnel

Answers kept only when the task verifies them



What it took

BIRD recipe: accelerator-hours by stage



4.2e18 training FLOPs

41.9 v6e chip-hours

5.1 teacher H200-hours

$136 at list prices

Recipe card

Student	Qwen3-1.7B, non-thinking
Teacher	DeepSeek-V4.1-Flash
Adapter	LoRA r256 · alpha 512 · 7 projections
SFT	lr 1e-4 · batch 256 · 3 epochs
Sequence	≤ 8,192 tokens · 2k/4k/8k buckets
GRPO	G = 8 · clip 0.2 · beta 0.02
GRPO lr	2e-6, then 5e-6 · ≤ 256 steps/round
Mesh	FSDP 8 × TP 1 on v6e-8
Serving	vLLM-TPU · TP 1 · bf16 KV
Stack	Tunix · qwix · Flax · Optax · JAX

Find the recipe on:

GitHub